Schrödinger's Toolbox: Exploring the Quantum Rowhammer Attack

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INTRODUCTION

Quantum computing remains in the noisy intermediate-scale quantum (NISQ) era, but rapid progress is bringing practical applications and transformative potential increasingly within reach [1–3]. As quantum computers move toward real-world deployment and begin interacting with sensitive systems, their security becomes a pressing concern [4].

Further, quantum circuits—from high-level algorithms to their compiled, hardware-specific instructions—constitute valuable intellectual property. As the field matures, protecting the integrity and confidentiality of these circuits will be critical to ensuring the trustworthiness and competitiveness of quantum platforms.

Even in their early stages, quantum systems have already shown vulnerability to a range of attack vectors. Fault-injection attacks, for instance, can disrupt quantum operations through physical manipulation of the quantum processor, leading to corrupted computations or unintended leakage of information [5]. Additionally, side-channel attacks have demonstrated the ability to reverse-engineer quantum programs by analyzing power consumption patterns during execution, exposing sensitive algorithmic structure and control flow [6].

Despite differences in implementation, quantum systems face analogous attack vectors to those in classical computing, highlighting the need to adapt classical security frameworks to emerging quantum technologies. Among these threats, hardware-level integrity violations warrant particular attention, as they often bypass conventional software-based defenses and can corrupt computations at the physical layer.

For instance, quantum hardware faces a unique threat known as *cross-talk*—a phenomenon where operations on one qubit inadvertently influence adjacent qubits due to residual interactions or electromagnetic coupling [7, 8]. These unintended interactions can corrupt qubit states, degrade circuit fidelity, and introduce non-local errors that are especially difficult to detect and correct. As devices scale and gate densities increase, mitigating cross-talk becomes critical to preserving the reliability and correctness of quantum computations.

Although cross-talk is a familiar source of noise, recent research shows it can also serve as a deliberate attack channel [9, 10]. Tan *et al.* demonstrated a fault-injection technique that leverages cross-talk on IBM's cloud-hosted superconducting quantum processors [10]. Remarkably, the attack can be executed with standard user privileges, yet still critically disrupts the behavior of neighboring qubits.

However, because their method relies on Qiskit Pulse—a low-level control interface that IBM has since begun deprecating [11]—its applicability to future devices may be limited. This deprecation may be motivated, at least in part, by the security risks of granting public users direct pulse-level access to the hardware.

Yet, as this work and others demonstrate, low-level control is not strictly necessary to launch an integrity attack that exploits inter-qubit interactions. Even using only high-level circuit operations, repeatedly applying gates to neighboring qubits can induce faults through residual cross-talk—closely paralleling the classical *rowhammer* attack on DRAM. In that context, rapid and repeated access to certain memory rows causes charge to leak into adjacent "victim" rows, resulting in unintended bit-flips [12, 13]. Critically, these faults occur without directly accessing the corrupted data, enabling silent and indirect manipulation.

Analogously, driving a "control" qubit with an especially dense sequence of gates can leak errors into neighboring qubits via cross-talk, degrading the fidelity of a victim circuit much as Rowhammer degrades DRAM. Almaguer-Ángeles *et al.* recently showed that this Quantum Rowhammer attack works even when the attacker is limited to Clifford gates—the standard 1- and 2-qubit operations such as X, H, S, CNOT and CZ [9].

Clifford gates are not an optional, low-level feature like Qiskit Pulse; they form the backbone of every contemporary quantum programming model. These operations underpin state preparation, basis transformations, entanglement generation, and the syndrome extraction procedures used in quantum error correction [14].

Because they are essential and universally accessible in cloud-based platforms, hardware must expose them without restriction. This unavoidably leaves systems vulnerable to cross-talk-based fault-injection attacks that exploit the native gate set itself.

In this work, I first replicate the experiment of Almaguer-Ángeles *et al.* and confirm that a minimal gate set—comprising only the single-qubit X gate and the two-qubit controlled-NOT (CNOT)—is sufficient to mount a successful Quantum Rowhammer (QR) attack. No access to pulse-level controls or non-Clifford operations is required; a carefully timed burst of CNOTs on aggressor qubits reliably induces bit-flip errors in an adjacent victim, reproducing the fidelity loss reported in the original study.

To probe the nature of the induced errors, I repeated the attack in the X-basis by applying Hadamard gates before and after measurement. The resulting bit-flip rates clustered tightly around 50% across all target qubits—independent of location or initial state—suggesting substantial phase randomization. In contrast, Z-basis outcomes exhibited more variability, particularly across different initial states. These results indicate that the Quantum Rowhammer attack likely introduces phase-dominated errors through residual coupling, demonstrating its ability to degrade quantum computations even without direct access to the affected qubits.

To reduce the damage caused by QR, I investigated a series of mitigation strategies. The first approach was dynamical decoupling (DD)—a widely used pulse-level technique that suppresses decoherence by inserting carefully timed π-pulses (typically X or Y gates) during idle periods in a quantum circuit [15]. These pulses effectively refocus the qubit's phase, averaging out low-frequency noise and mitigating both dephasing and certain crosstalk effects, all without altering the intended logic of the circuit [16].

Common DD strategies include the Uhrig DD and uniform DD, which have been shown to extend qubit coherence times—especially T_2—by up to an order of magnitude in both superconducting and trapped-ion platforms [17–19]. Since DD sequences can be applied entirely through circuit-level compilation (e.g., idle padding), they represent a practical, software-based defense against QR-style phase errors that does not require modifications to the underlying hardware.

Finally, I implement a simple error detection and post-selection scheme to mitigate QR-induced faults. This method encodes the target qubit using a 3-qubit repetition code: three ancilla qubits are entangled with the data qubit via CNOT gates, and the circuit outcome is retained only if a majority vote of the ancilla measurements matches the expected initial state. Such repetition codes offer lightweight protection against bit-flip errors and have been widely used in both experimental and theoretical quantum error correction research [20, 21]. Post-selection enhances fidelity by discarding runs where an error is detected, thereby suppressing error propagation at the cost of reduced sampling efficiency [22].

In summary, I (i) demonstrate that a minimal Clifford-only gate set is sufficient to launch a Quantum Rowhammer attack on today's superconducting hardware; (ii) explore the impact of the QR attack in the X-basis; (iii) evaluate the spatial and temporal reach of the QR attack; and (iv) evaluate two compilation-level defenses—dynamical-decoupling, and a lightweight three-qubit error detection code. Together, these results reveal subtle but practical integrity vulnerabilities in current cloud-accessible quantum hardware and outline practical near-term mitigation strategies.

BACKGROUND

Quantum Computing Fundamentals

While a classical bit must be either 0 or 1, a quantum bit—or *qubit*—can exist in a superposition of both states simultaneously. Formally, the state of a qubit $|\psi\rangle$ is described as a linear combination of two orthonormal basis states, $|0\rangle$ and $|1\rangle$, with complex coefficients:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle, \quad \text{where } \alpha, \beta \in \mathbb{C},\ |\alpha|^2 + |\beta|^2 = 1.$$

This normalization ensures that a measurement of the qubit in the computational basis yields either $|0\rangle$ with probability $|\alpha|^2$ or $|1\rangle$ with probability $|\beta|^2$.

Quantum operations are implemented via *unitary gates*—linear transformations U that preserve the norm of the state vector. These gates satisfy $U^\dagger U = UU^\dagger = I$, where $U^\dagger$ is the Hermitian adjoint of U. Some of the most fundamental single- and two-qubit gates include:

- **Pauli-X gate:**
$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad X|0\rangle = |1\rangle,\ X|1\rangle = |0\rangle$$
The X gate acts as a quantum analog of the classical NOT gate, flipping $|0\rangle$ to $|1\rangle$ and vice versa.

- **Hadamard (H) gate:**
$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad H|0\rangle = |+\rangle,\ H|1\rangle = |-\rangle$$
The Hadamard gate creates superposition, mapping basis states to the *X-basis*: $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, $|-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$.

- **Controlled-NOT (CNOT or CX) gate:**
$$\text{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$
The CNOT gate operates on two qubits: if the *control* qubit is in state $|1\rangle$, it flips the *target* qubit; otherwise, it leaves the target unchanged. CNOT gates are essential for generating entanglement, which has no classical analog.

Quantum Circuits

Quantum circuits are built by sequencing quantum gates across multiple qubits over time. Each line in a circuit diagram corresponds to a qubit, and horizontal connections represent gate operations. The resulting computation is unitary and reversible (until measurement), and the overall behavior of a quantum program is governed by the coherent interference of amplitudes across all computational paths.

After a quantum circuit is constructed at the logical level, a quantum development kit—such as IBM's Qiskit—compiles it into a form compatible with the target hardware [23]. This compilation process, known as transpilation, adapts the high-level gate sequence to the device's native gate set and physical qubit topology. Logical operations are decomposed into basis gates, and any required routing (e.g., for non-adjacent qubit interactions) is inserted automatically.

During execution, these gate instructions are ultimately implemented as calibrated microwave pulse sequences that manipulate the physical qubits. Once the circuit completes, the qubits are measured in the computational basis.

Because NISQ devices are noisy and non-deterministic, each circuit is executed many times—typically thousands—to estimate outcome probabilities. Each run is called a shot, and the final output distribution reflects the empirical frequencies of different measurement results across all shots.

NISQ Computers

Noisy Intermediate-Scale Quantum (NISQ) devices typically feature between 10 and 100 qubits and are subject to high error rates from sources such as decoherence, gate infidelity, readout noise, and cross-talk [2]. Despite these limitations, NISQ devices can perform certain tasks that challenge classical computers. Fig 1 shows the topology of IBM's 127-qubit Eagle r3 processor, which arranges superconducting transmon qubits in a heavy-hexagonal lattice, where each qubit connects to one to three neighbors [23].

Quantum-as-a-Service and Multi-Tenancy

Most commercial providers—including IBM Quantum, Amazon Braket, and Microsoft Azure Quantum—offer Quantum-as-a-Service (QaaS): remote access to real hardware through web APIs and Python SDKs. As demand grows, vendors are increasingly exploring *multi-tenant* scheduling, wherein circuits from different customers would share a single processor either time-sliced (sequential jobs) or co-resident (simultaneous execution on disjoint qubit subsets) [24].

While multi-tenancy is not yet available on current devices, its adoption is anticipated as hardware scales. Multi-tenancy promises higher utilization and reduced queue latency, but also raises new security questions: physical noise processes—especially cross-talk—do not respect logical job boundaries, meaning an adversarial workload can, in principle, perturb neighboring circuits that occupy nearby qubits in space or time. This vulnerability motivates the Quantum Rowhammer threat explored in the remainder of this paper.

THREAT MODEL

This work considers a threat model motivated by emerging trends in quantum cloud computing, particularly the shift toward scalable, shared hardware. While today's superconducting quantum processors (e.g., IBM's 127-qubit heavy-hex architecture in Fig. 1) do not yet support true multi-tenant execution—where circuits from multiple users run concurrently on shared hardware—this model is anticipated as quantum demand increases [24].

Here, I explore the security implications of co-located circuit execution, drawing parallels to classical cloud co-residency attacks. The adversary is assumed to be an ordinary cloud user (user A) with no privileged access, physical control, or visibility into other users' jobs. Their capabilities are limited to submitting arbitrary quantum circuits composed of the provider's native gate set (e.g., Clifford operations and standard single- and two-qubit gates) via the standard API. The victim (user B) is a co-resident tenant scheduled to run circuits on adjacent qubits at overlapping.

The attacker's goal is to degrade the fidelity of the victim's computation while submitting a syntactically valid and policy-compliant quantum circuit. The attack leverages physical cross-talk—unintended residual coupling between neighboring qubits—to inject faults indirectly. By applying dense sequences of native gates (e.g., X, CNOT) to "aggressor" qubits, the attacker induces decoherence and stochastic errors on adjacent "victim" qubits.

ATTACK SCENARIOS

Scenario A: Co-Resident Sabotage

A victim job and an attacker job are loaded onto the device during the same time, in adjacent portions of the lattice. The attacker requests an innocuous-looking circuit that performs an intense burst of CNOTs on its allocated qubits—the *single-center QR* experiment. During the victim's computation, cross-talk flips the victim's state, distorting results without leaving a classical trace.

Scenario B: Wide-Sweep Denial of Coherence

A determined adversary can submit a series of QR jobs, each targeting a different set of control qubits, thereby sweeping the fault-inducing region across the chip to maximize coverage and disruption. This scenario mirrors classical rowhammer amplification techniques—such as Sledgehammer—which induce widespread corruption by simultaneously hammering multiple memory regions, thereby escalating the impact beyond isolated bit-flips [25].

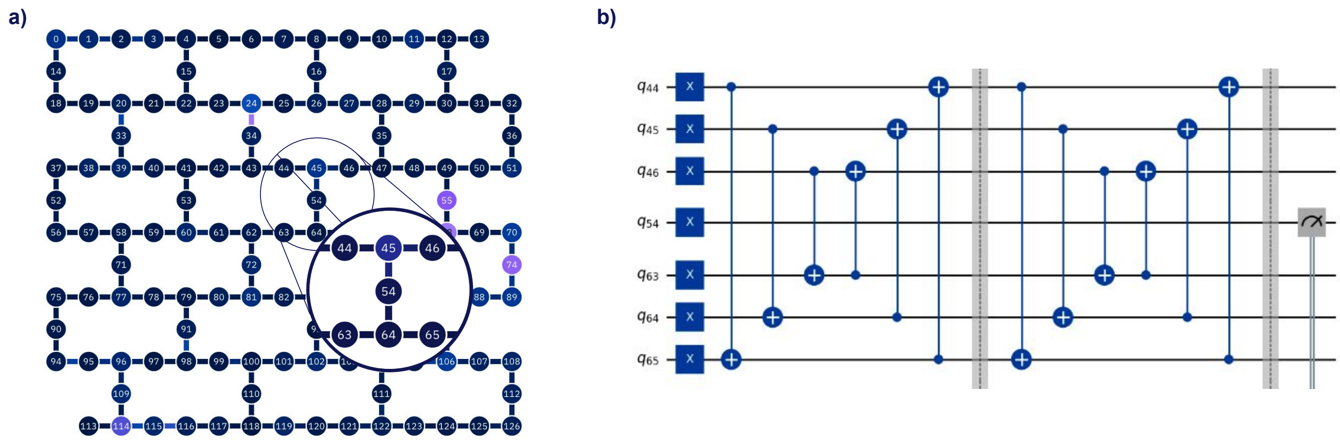

FIG. 1: Targeted Quantum Rowhammer Circuit. (a) Hardware coupling map of IBM's 127-qubit Eagle device. The "neighborhood" for qubit 54 is enlarged, illustrating the control-target layout for the Quantum Rowhammer experiments. **(b)** Example circuit schematic for the attack on qubit 54. The center qubit is initialized in the $|1\rangle$ state and repeatedly hammered via multiple rounds of pairwise CNOT gates between its neighboring qubits.

METHODOLOGY

This section describes the experimental framework used to (i) launch the Quantum Rowhammer (QR) attack, (ii) sweep the attack across an entire heavy-hex lattice, and (iii) evaluate two software-level countermeasures—dynamical decoupling (DD) and a three-qubit repetition code with post-selection. All experiments were executed on cloud-accessible superconducting backends (IBM *Brisbane* and *Sherbrooke*) with 127 physical qubits and the provider's default transpiler.

For any chosen *center* qubit, I enumerated its six nearest neighbors in the heavy-hex coupling graph (two direct bus connections plus their immediate upstream and downstream partners) (See Fig. 1a). This six-qubit square is referred to as the *hammer neighborhood*.

Attack Protocols

Single-Target

The hammer attack has five stages (Fig. 1b):

1. **Initialization:** The center qubit is prepared in either $|0\rangle$ or $|1\rangle$ (or, for the X-basis study, in $|+\rangle$ or $|-\rangle$ via a Hadamard pre-rotation).
2. **Neighbor burst:** An X-gate is applied to the six neighbors.
3. **Pair-wise entangling rounds:** The neighbors are then entangled with one another through $CNOT$ gates executed in symmetric pairs, repeated for a fixed number of rounds. This concentrates residual ZZ interactions on the idle center qubit.
4. **Basis undo (optional):** If the qubit was shifted to the $|\pm\rangle$ basis, a second Hadamard is applied to return to the computational basis.
5. **Measurement:** The center qubit is measured, and the frequency of logical flips is recorded over 20,000–40,000 shots.

Lattice-Wide Sweep

To study spatial propagation, I extended the hammer protocol to multiple, disjoint center sets. In each cycle a new group of centers was hammered in parallel while every qubit on the chip was measured. The experiment interleaved hammer and benign cycles to serve as an internal control and assess whether fidelity loss in the hammered qubit persisted over time.

Mitigation Techniques

To counteract the effects of the QR attack, I evaluate two circuit-level mitigation strategies that require no hardware modifications: (1) dynamical decoupling (DD), and (2) a simple repetition-based post-selection scheme. Both are applied via transpilation and classical post-processing, making them practical for deployment in current cloud-accessible quantum platforms.

Dynamical decoupling.

Dynamical decoupling (DD) combats decoherence and certain cross-talk effects by applying carefully timed π-pulses during idle periods in the circuit. This work implements two DD strategies:

- **Uniform DD**: Each idle window is padded with an X–I–X sequence, equally spaced across the idle duration.
- **Uhrig DD**: A higher-order sequence that places n π-pulses at non-uniform time intervals.

These DD layers are inserted using a transpiler pass that performs As-Late-As-Possible scheduling to expose idle windows, then pads each window with the appropriate pulse sequence. The quantum logic of the original circuit is left unchanged.

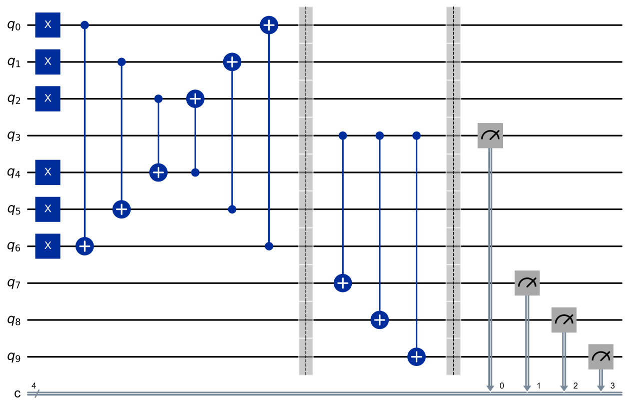

FIG. 2: Post-selection-based error detection circuit. The data qubit (q_3) is entangled with ancillas (q_0, q_1, q_2) via CNOT gates following the hammer circuit. The result is kept only if a majority vote among the ancillas matches the data qubit's initial state.

Repetition-based post-selection.

As a second mitigation strategy, I implement a basic fault-detection scheme using three ancilla qubits and classical post-selection (Fig 2). The circuit structure is as follows:

1. The central data qubit is prepared in either the $|0\rangle$ or $|1\rangle$ state.
2. The hammer neighborhood is driven with X and repeated CNOT gates in a cross-talk-inducing pattern designed to inject noise into the data qubit.
3. Three ancilla qubits are entangled with the data qubit via CNOT gates, with the data qubit as control and each ancilla as target.
4. All four qubits are measured.
5. A *majority vote* is taken over the three ancilla measurements. If the vote matches the expected initial state, the result is retained; otherwise, the shot is discarded.

This scheme rejects erroneous results likely induced by cross-talk during the hammering phase, thereby increasing measurement fidelity. While this approach reduces the number of usable samples, it requires no real-time correction or additional quantum hardware and is fully compatible with current NISQ devices.

RESULTS

Single-Target Quantum Rowhammer

To evaluate the vulnerability of individual qubits to the QR attack, I conducted a targeted fault-injection experiment. For each trial, a designated *center* qubit was initialized in either the $|0\rangle$ or $|1\rangle$ state and subjected to a QR circuit composed of aggressive gate operations on its neighbors. A corresponding baseline circuit, omitting the hammering phase, was also executed for comparison. The flip probability—defined as the likelihood that the measurement outcome deviates from the initial state—was recorded for each configuration.

Figure 3 shows the bit-flip probabilities across a selection of center qubits, stratified by initial state. In all cases, the QR protocol significantly increased the flip probability relative to the baseline. For initial state $|0\rangle$ (left panel), flip probabilities under hammering consistently rose above 40%, with some cases exceeding 50%. In contrast, the corresponding baseline error rates remained below 15%, and often below 5%.

The effect is similarly pronounced for initial state $|1\rangle$ (right panel), though with slightly higher variability across qubits. Notably, center qubit 109 exhibits a flip probability of over 70 % under attack, highlighting the location-dependent nature of cross-talk-induced faults.

Across all center qubits and both backends, circuits initialized in the $|1\rangle$ state exhibited higher average bit-flip rates than those initialized in $|0\rangle$. Specifically, the average flip probability for $|0\rangle$ was approximately 0.46, whereas $|1\rangle$ yielded a higher rate of 0.58. This discrepancy aligns with physical intuition: in superconducting qubits, spontaneous relaxation from the excited state $|1\rangle$ to the ground state $|0\rangle$ (i.e., T_1 decay) occurs more readily than excitation in the reverse direction. As a result, circuits beginning in $|1\rangle$ are more susceptible to bit-flip errors under cross-talk–induced perturbations.

Hadamard Basis

Figure 4 shows the impact of QR in the X-basis, where each target qubit is first rotated via a Hadamard gate. When the hammer sequence is applied, flip probabilities converge tightly around 50% (mean $p\_\mathrm{flip} = 0.498$, $\sigma < 0.013$) regardless of qubit index or initial state, indicating strong phase randomization consistent with cross-talk-induced dephasing. In contrast, control runs without hammering yield negligible flips (mean $p\_\mathrm{flip} = 0.027$), confirming the attack's effect.

Table I compares these results with those from the Z basis. While Z-basis flip rates vary widely depending on center qubit and initial state (ranging from 0.34 to 0.84), X-basis flip rates are tightly centered around 0.5. This contrast supports the hypothesis that QR predominantly introduces *phase errors*, which convert into deterministic bit-flips in the conjugate basis.

TABLE I: Average bit-flip probabilities under hammering in the computational (Z) and Hadamard-shifted (X) bases.

Basis	Hammered state	p_{flip}
Z	0	0.455 ± 0.080
Z	1	0.592 ± 0.130
X	0/1	0.498 ± 0.013

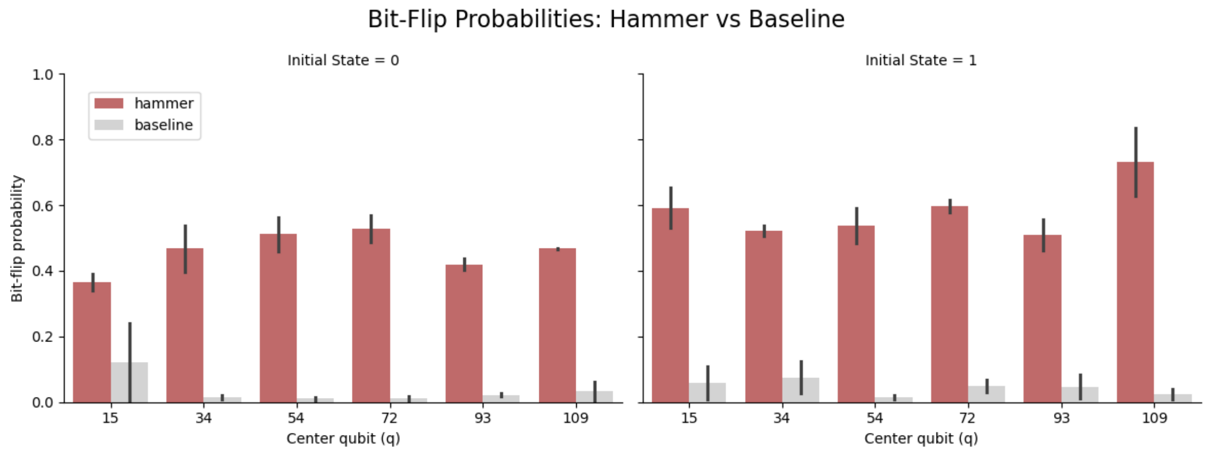

FIG. 3: Bit-flip probabilities for individual target qubits under baseline conditions and Quantum Rowhammer (QR) attack. Each subplot shows results for a different initial state ($|0\rangle$ left, $|1\rangle$ right). Red bars indicate the flip probability after aggressive gate activity (hammering) on neighboring qubits, while gray bars show the baseline flip probability in the absence of such activity. Error bars denote standard error across multiple trials. QR consistently induces significantly higher error rates, demonstrating its effectiveness as a fault-injection mechanism.

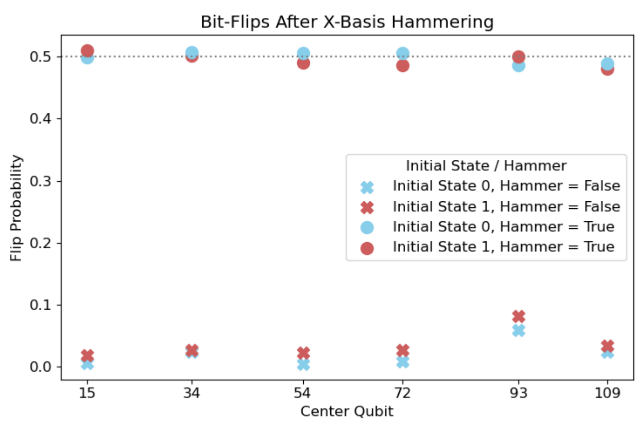

FIG. 4: Bit-flip probability after Hadamard-shifted hammering. Each point is the bit-flip likelihood for a single center qubit. Crosses are control runs without hammering; dots include the hammer burst. Hammering drives the flip-probability to $\approx 50\%$ for both initial states, whereas control runs stay below 3%.

Specifically, qubits initialized in the X-basis are sensitive to phase flips (i.e. $|+\rangle \xrightarrow{\text{Z error}} |-\rangle$, and vice versa). Thus, frequent phase disturbances, such as Z-errors and ZZ-interactions, effectively randomize the state of a qubit in the X-basis. When the qubit is subsequently rotated back to the Z-basis via a Hadamard gate, this phase randomization manifests as a uniform probability of measuring $|0\rangle$ or $|1\rangle$. The consistency of this effect across initial states and central qubits suggests that the QR attack primarily induces dephasing.

Rowhammer Sweep Across the Lattice

Figure 5 tracks the bit-flip probability of all 127 physical qubits over forty alternating cycles of (i) an aggressive, multi-center hammer burst and (ii) a benign reference cycle. Several trends emerge from this sweep.

First, the corruption is local but not global. During each hammer cycle (even-numbered columns), qubits within or directly coupled to the targeted center group exhibit flip probabilities as high as $p_{\text{flip}} \approx 0.8$ (bright teal). The disturbance typically extends one or two coupling-graph hops beyond the driven neighborhood but does not propagate coherently across the full lattice.

Second, there is rapid temporal recovery. In nearly all cases, the subsequent benign cycle (odd-numbered columns) returns the affected qubits to the background error floor ($p_{\text{flip}} < 0.05$, black). This suggests that the cross-talk burst impairs victim computations momentarily, as the damage does not persist in future runs.

Third, the attack is cycle-to-cycle consistent. The spatial envelope of elevated error rates closely tracks the scheduled center group, demonstrating that the disturbance is deterministic and reproducible—not a byproduct of random device drift.

Together, these findings indicate that Quantum Rowhammer can be weaponized to disrupt neighboring workloads in a time-sliced, cloud-based setting. However, its effects appear to be temporally bounded, largely confined to the execution window of the malicious job.

Error-mitigation results

Figure 6 summarizes the effectiveness of the two software–level defenses evaluated against QR.

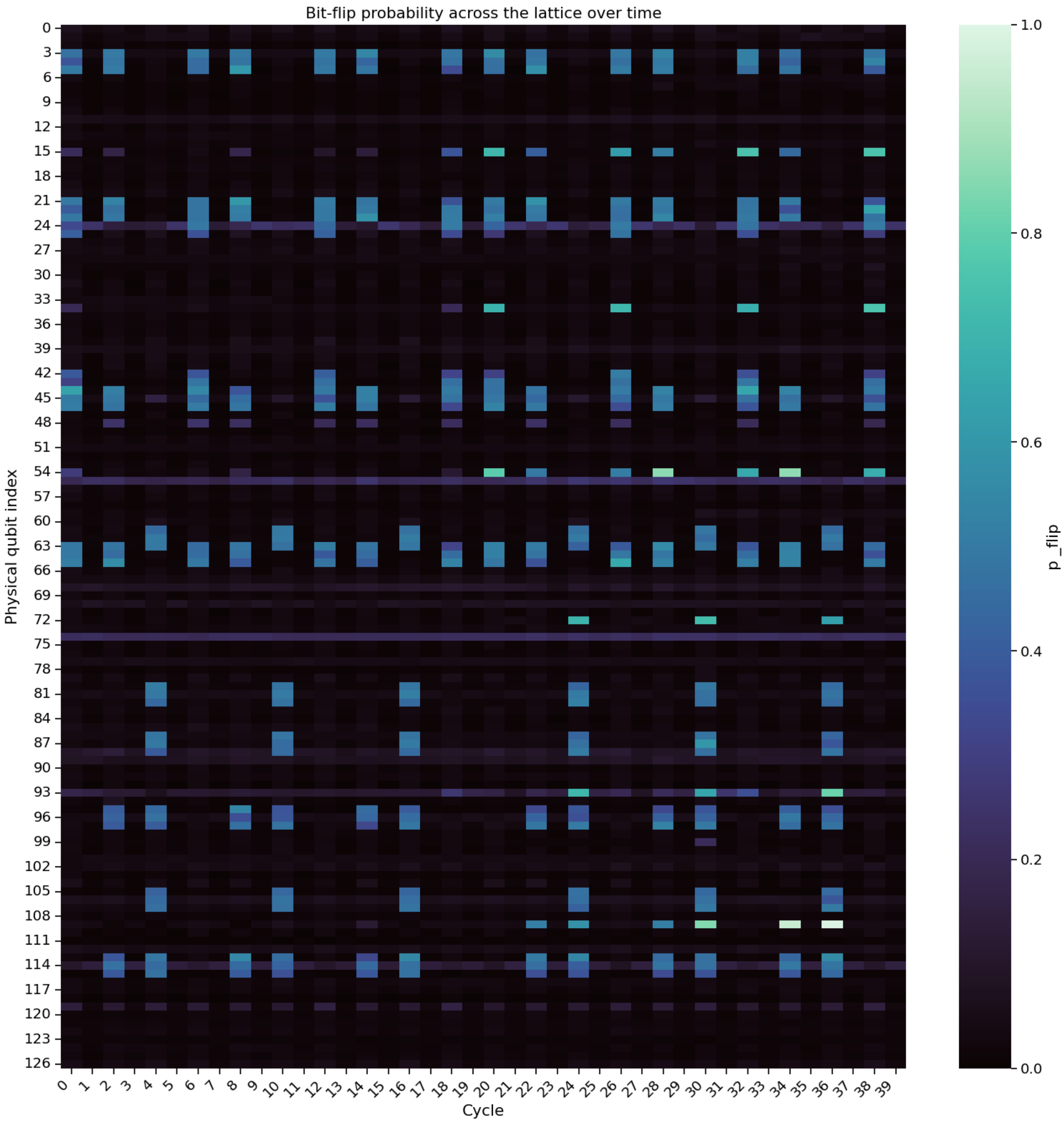

FIG. 5: Lateral and temporal spread of Quantum Rowhammer. The heat-map shows the measured bit-flip probability p_{flip} for every physical qubit (y-axis) over 40 execution cycles (x-axis) on `ibm_brisbane`. Even-numbered cycles contain a multi-center hammer burst; odd cycles run a benign circuit. Bright pixels reveal qubits whose fidelity is strongly degraded by the attack; dark pixels denote baseline error rates below 0.05. High-error regions appear only during hammer cycles and remain confined to qubits that are at most two hops from the driven centers, while the device recovers to nominal behaviour in the immediately following benign round.

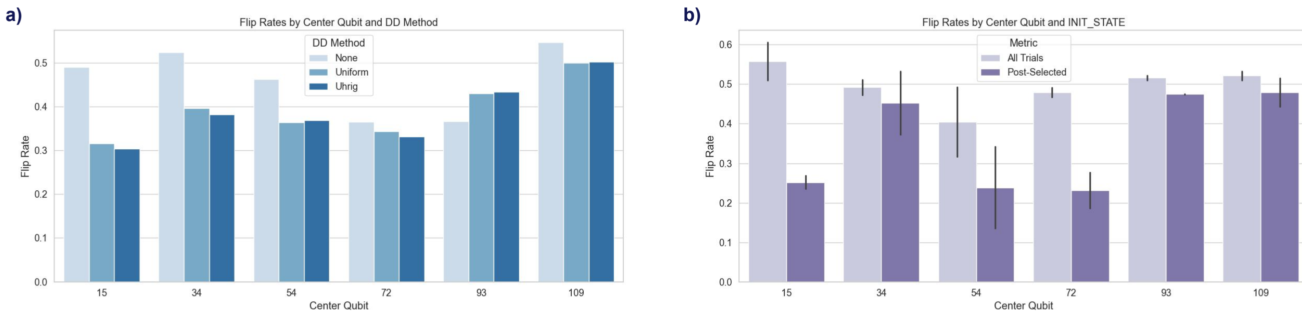

FIG. 6: Effectiveness of software-level mitigations. (a) Bit-flip probabilities with no protection, uniform dynamical decoupling, and Uhrig DD on six representative center qubits. **(b)** Flip probabilities for the same centers when a three-qubit repetition code is employed; light bars show all shots, dark bars only the post-selected subset that passes the majority-vote check. Error bars denote the $\pm 1\sigma$ binomial confidence interval.

Dynamical decoupling. Padding the victim circuit with idle-time X pulses provides a measurable but modest benefit. Both uniform spacing and the Uhrig sequence lower the average bit-flip probability by $\sim 20\%$ relative to an unprotected circuit (mean reduction $\Delta p_{\text{flip}} = 0.11$ across all centers). Improvements are most pronounced on qubits that already suffer only moderate disturbance (e.g. centers 15 and 34), whereas heavily affected sites such as qubit 109 remain above $p_{\text{flip}} \approx 0.50$. The two DD schedules perform nearly identically, suggesting that the dominant cross-talk noise is broadband rather than strongly frequency-dependent.

Repetition code with post-selection. Adding a three-qubit repetition code and discarding shots whose ancilla majority vote signals an error further suppresses the observed bit-flips. Across all centers, the retained shots exhibit a $36\% \pm 7\%$ lower error rate than the unfiltered set. The gain is particularly large on the previously vulnerable centers 15 and 72, cutting their flip probabilities to 0.26–0.28. The downside is efficiency: on average only 58% of shots survive post-selection, so throughput drops proportionally. Nonetheless, when high fidelity is required, this scheme offers a simple, compiler-level defense that does not rely on hardware changes or pulse-level access.

Taken together, the results indicate that *compiler-inserted* mitigations—while unable to eliminate Quantum Rowhammer's effects—can meaningfully raise the attacker's costs.

DISCUSSION

Physical origin and spatial footprint. The data show that an otherwise innocuous burst of native Clifford operations can inject errors into adjoining qubits with remarkable determinism. As shown in Fig. 5, the elevated flip probabilities (i.e. $p_{\text{flip}} \geq 0.35$) appear exclusively during the even-numbered hammer cycles, forming distinct vertical bands that correspond to the locations of actively driven center qubits.

These high-error regions align closely with the device's heavy-hex coupling topology, extending no more than one or two hops from the targeted sites. This spatial correlation suggests that residual ZZ-type interactions along shared bus resonators are the primary pathways for noise propagation [26].

Notably, the effect is both localized and ephemeral. In nearly every case, qubits affected during the hammer phase revert to baseline error rates in the immediately following benign cycle. This rapid recovery suggests that the induced noise is not due to lasting calibration drift or thermal damage, but instead behaves like a transient, controllable phase-randomization process—one that can be triggered and silenced at will.

Such precise temporal control yields a nuanced attack profile: the injected faults are both spatially targeted and temporally bounded, making them difficult to detect and highly repeatable. This enhances the attack's subtlety, but also constrains its scope—its effects are short-lived and confined to specific windows of execution.

Effectiveness of compiler-level defenses. Implementing dynamical decoupling (DD) and repetition-code post-selection has shown measurable reductions in error rates. However, neither approach fully neutralizes the threat posed by cross-talk-induced errors. While DD sequences can mitigate certain types of decoherence, their effectiveness varies depending on the specific qubit interactions and environmental conditions [15]. Moreover, the combination of DD and post-selection introduces trade-offs, such as increased execution times and reduced data throughput, which may not be acceptable in all operational contexts

Given these limitations, relying solely on software-level defenses is insufficient. Hardware-based solutions, such as improved qubit isolation and real-time cross-talk monitoring, may be necessary to achieve robust protection against such vulnerabilities [10].

Implications for multi-tenant processors. While current cloud schedulers time-slice jobs, larger devices in the near future are expected to place mutually unaware tenants on disjoint sub-lattices concurrently. These results show that spatial isolation is porous: a malicious tenant can degrade a neighbor's logical fidelity by up to $\approx 80\%$. More subtly, the attack need not strive for catastrophic failure—*predictable* noise is enough to bias expectation values in a variational algorithm or to derail a fragile error-mitigated computation. This vulnerability highlights the need for stringent isolation protocols and security measures in shared quantum computing environments [27].

Cross-talk as a covert channel. The lattice-sweep experiment also demonstrates that cross-talk can be exploited as a low-bandwidth *covert channel* in a manner reminiscent of the classical *Prime-and-Probe* cache attack. Here, the attacker (*sender*) first *primes* the device by issuing a QR burst on chosen center qubits; this saturates its immediate neighborhoods with phase noise. In the subsequent cycle the innocent tenant (*receiver*) *probes* the lattice by running a measurement-only circuit identical to the one used in Fig. 5.

The receiver classifies each qubit according to a simple threshold rule: if the measured bit-flip probability exceeds 0.35 the qubit is recorded as a logical '`1`'; otherwise it is recorded as '`0`'. Because only qubits near the hammered center exhibit elevated error rates, the receiver effectively "reads out" the attacker's signal via these correlated spatial patterns.

Information can be encoded in two complementary ways. First, the attacker can use the presence or absence of hammering to send a binary digit— hammering corresponds to '1', while skipping it corresponds to '0', forming a basic on–off keying scheme. Second, and more powerfully, the attacker can modulate *which* center qubit is hammered. If there are m disjoint neighborhoods that can be individually targeted without cross-talk overlap, then each hammer cycle can encode up to $\log_2 m$ bits, based on which neighborhood is activated.

For example, choosing among 8 candidate centers yields an additional 3 bits per cycle. Thus, each hammer cycle can carry up to $1 + \log_2 m$ bits of covert data. On current 127-qubit devices, it is feasible to identify a dozen or more well-separated hammer centers. This enables approximately $1 + \log_2 12 \approx 4.6$ bits of data per hammer–probe round.

At modest cycle times (e.g., 10 microseconds per round), this corresponds to a covert bandwidth of $\sim$460 bits per second—enough to exfiltrate cryptographic keys, variational parameters, or calibration secrets across tenant boundaries.

CONCLUSION

This work shows that even a minimal Clifford-only circuit is sufficient to mount a Quantum Rowhammer (QR) attack on modern superconducting qubit hardware. The induced faults are primarily phase errors, which manifest as near-maximal bit-flip rates when the victim qubit is probed in the conjugate basis. These errors are spatially localized—typically affecting qubits within one or two hops of the attacker—and temporally precise, emerging only during targeted hammer cycles. This combination of locality, repeatability, and stealth makes QR viable both as a sabotage tool and as a low-bandwidth covert communication channel.

While compiler-level countermeasures such as dynamical decoupling and post-selection provide meaningful mitigation, they fall short of fully neutralizing the threat. The attack's ability to inject transient, deterministic noise through standard gate operations underscores the need for deeper architectural defenses.

Defending against cross-talk–based attacks will likely require a layered strategy that spans hardware, middleware, and scheduling policies. On the hardware side, IBM has already begun deploying *tunable bus couplers*—inter-qubit links that can be dynamically turned off outside of gate operations—to suppress residual ZZ-interactions. The first production implementation appears in the 133-/156-qubit Heron family, which IBM reports has "practically zero cross-talk" [28, 29]. If the on-demand bus-isolation scheme proves robust, it represents a major step toward closing the side channel exploited by Quantum Rowhammer.

This study demonstrated the potential for a cross-talk channel in a controlled setting but did not build the full sender/receiver stack. A natural next step is to implement a proof-of-concept "Prime-and-Probe" protocol on today's cloud back-ends: the sender primes the lattice by hammering a chosen center and the receiver probes at the end of the cycle. After which, the receiver threshold-decodes the spatial error pattern, and streams the resulting bit string. Such an implementation would help quantify the real-world reliability, bandwidth, and stealthiness of this covert channel under realistic noise and scheduling conditions.

Ultimately, the Quantum Rowhammer attack highlights a deeper lesson for the future of secure quantum computing: as access to shared quantum hardware grows, so too does the risk of physical-layer interference between users.

Even standard gate primitives, when structured adversarially, can degrade nearby workloads or leak information across logical boundaries. As quantum devices scale and scheduling becomes increasingly parallelized, safeguarding against such side channels will be just as essential as improving qubit fidelity or algorithmic depth. The findings presented here underscore the need for quantum-aware security engineering at every level of the stack.

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